Maximum Homomorphism Density of Simple Graphs in Graphons

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By

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Abstract

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1. Introduction

Extremal combinatorics fundamentally asks the question about how local constraints affect global behavior. Historically, such questions were asked about graphs, which intuitively are collections of points and lines between them. Work in this area has led to fundamental theorems like Mantel's Theorem, Turán's Theorem, and the Kruskal-Katona Theorem [1]. However, the development of the theory of dense graph limits, specifically the introduction of graphons by Lovász and Szegedy [2], provided a powerful continuous framework for these discrete problems. Graphons act as limit objects for sequences of dense graphs, allowing us to study subgraph densities using the tools of real analysis, topology, and measure theory.

Philosophically, graphons represent the completion of the space of dense finite graphs, much as the real numbers complete the rationals [3]. In the discrete realm, graphs are notoriously rigid. For example, how do you compare two graphs of different sizes? Does the question asking how similar two finite graphs of different sizes are even make sense? Graphons wash away this discrete noise, capturing the pure "structural essence" of a network. By mapping graphs to symmetric measurable functions, the fundamentally combinatorial act of counting subgraphs is transformed into the analytic act of integration.

This paradigm shift has revolutionized multiple areas of discrete mathematics. Graphons have provided the natural language to unify theories of quasi-randomness [4], formulate continuous analogues of the Szemerédi Regularity Lemma [5], and develop robust property testing algorithms for massive networks [6]. Beyond pure mathematics, they serve as foundational models in network science [3] and machine learning [7], offering a way to study the large-scale behavior of the internet, social networks, and biological systems where individual nodes are ephemeral, but the underlying continuous probability distributions governing their connections remain stable.

A central object of study is the Razborov Triangle $\mathcal{R}$, which characterizes the exact region of simultaneously achievable edge and triangle densities for any graphon. While the exact boundary of $\mathcal{R}$ is known [8], the behavior of other subgraph densities within this region remains a rich area of inquiry. This thesis focuses on a specific, largely unexplored extremal question: if we fix an edge density ε and force the triangle density τ to be infinitesimally small, how much of an arbitrary simple graph G can survive? To make this precise, let $T^G_{\max}(\varepsilon, \tau)$ denote the maximum possible homomorphism density of G over all graphons with edge density ε and triangle density at most τ.

To build intuition, one can view this problem through the physical lens of statistical mechanics and complex systems. If we consider vertices as individual entities or particles, the edge density ϵ measures the global frequency of simple 2-particle interactions. The triangle density τ, meanwhile, measures the fundamental 3-particle interactions. From this perspective, understanding $T^G_{\max}(\varepsilon, \tau)$ as $\tau \to 0$ is fundamentally a question about the hierarchical breakdown of complexity: if we fix the global rate of 2-particle interactions but actively suppress the 3-particle interactions, how quickly are more complex, higher-order multi-particle interactions

(represented by the arbitrary graph G) forced to die out? The graph parameter $\alpha(G)$ developed in this thesis quantifies exactly this rate of decay.

The main result of this thesis is that this decay is governed by a clean, power-law scaling, and that the exponent of this scaling is inherently tied to the structural relationship between the graph G and its triangles. capture this, we introduce a novel graph parameter, denoted $\alpha(G)$. This parameter is exactly calculated via a linear program whose variables correspond to the vertices and edges of G, and whose constraints are dictated by the triangles within G.

Theorem 1.1 *(Theorem 7.4)* For any graph G containing at least 1 triangle where no edge connects a vertex to itself, and for any fixed edge density $\varepsilon \in \left(0, \frac{1}{2}\right]$, the maximum homomorphism density of G scales asymptotically as

$$T^G_{\max}(\varepsilon, \tau) = \Theta\left(\tau^{\alpha(G)}\right) \tag{1}$$

as the triangle density $\tau \to 0$.

To establish this result, we utilize a dual approach. The upper bound, $T^G_{\max}(\varepsilon, \tau) = O\left(\tau^{\alpha(G)}\right)$, is proven by decomposing arbitrary graphs into a collection of simpler graphs. We then apply Finner's Generalized Hölder inequality [9] in conjunction with Strong Duality in Linear Programming [10]. to tightly bound the homomorphism density. To prove the bound is sharp, yielding the matching lower bound, $T^G_{\max}(\varepsilon, \tau) = \Omega\left(\tau^{\alpha(G)}\right)$, we explicitly construct a family of graphons. This construction is parametrically guided by the optimal solutions to the linear program, anchoring our graphons tightly to the τ boundary.

The remainder of this thesis is organized as follows. Section 2 collects the necessary preliminaries, defining graphons, cut metrics, homomorphism densities, and the relevant tools from linear programming and measure theory. Section 3 formally defines the extremal problem. Section 4 introduces the Triangle Spanning Decomposition, a structural tool necessary for our upper bounds. Section 5 formally defines the linear program that yields the graph parameter $\alpha(G)$ and explores its dual properties. Section 6 proves the upper bound for triangle-spanning graphs, which is then generalized to all simple graphs in Section 7, alongside the matching lower-bound constructions to prove our main theorem. In Section 8, we exhibit example graphs, their associated values of $\alpha(G)$, and graphon constructions which produce the desired behavior. Finally, we conclude in Section 9 by discussing the implications of these bounds, open questions regarding the implicit constants, and connections to the Szemerédi-Ruzsa Triangle Removal Lemma.

2. Preliminaries

This section goes over the necessary background needed for this thesis.

2.1. Graphons

This section establishes the foundational machinery of graphon theory that will be used throughout our analysis. For comprehensive proofs and a broader historical context, we direct the reader to Chatterjee's monograph [11] and Lovász's textbook [3]. However, before passing to these continuous limit objects, we must briefly set our standard graph-theoretic terminology.

Definition 2.1 A **simple graph**, informally, is a collection of points together with lines connecting the points, where no line can connect a point to itself. Formally, we define a simple graph G as a pair $G = (V, E)$ where V is finite set, identified with $\{1, ..., |V|\}$, and E is a set of unordered pairs of distinct elements of V. We call elements of V vertices and elements of E edges. When we need to refer to the vertex and edge sets of a particular graph G, we can write $V(G)$ and $E(G)$. Let the set of simple graphs be $\mathcal{G}$.

Graphons were introduced by Lovász and Szegedy [2] as a limit object for sequences of dense simple graphs. Before giving the formal definition, we motivate the choice of object by describing how a graphon parametrizes a natural random graph model — a perspective that will make the equivalence relations in the formal definition feel inevitable rather than arbitrary.

Given a symmetric measurable function $W : [0, 1]^2 \to [0, 1]$ and an integer n, the ***W*-random graph** $G(n, W)$ on a vertex set $\{1, ..., n\}$ is constructed in two steps. First, independently sample n random variables $U_1, ..., U_n$ uniformly on $[0, 1]$ and assign the label U_i to vertex i. Secondly, independently on each pair $i < j$, include the edge $\{i, j\}$ with probability $W(U_i, U_j)$. The value $W(x, y)$ is then the connection probability between vertices labeled x and y and the uniform distribution on $[0, 1]$ is the distribution over the labels. This construction recovers the Erdös–Rényi model $G(n, p)$ when $W \equiv p$ is constant, and recovers the stochastic block model when W is a step function, which we will call multipodal.

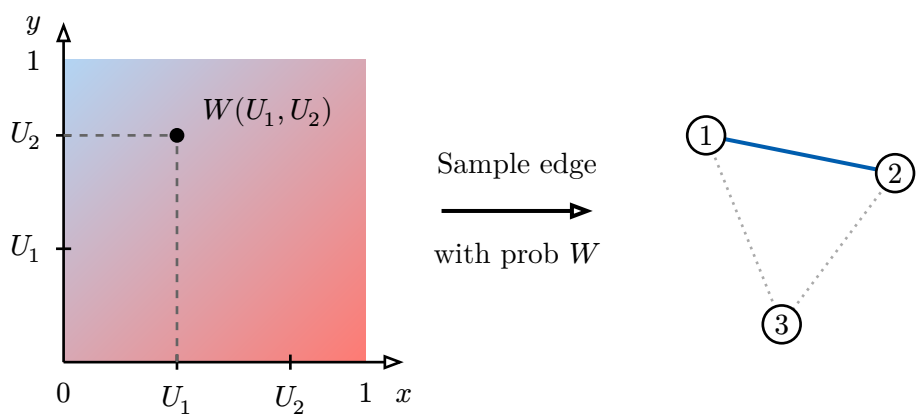

Figure 1: The W-random graph sampling process. Vertices are assigned uniform random labels $U_i \in [0, 1]$. An edge between vertices i and j is subsequently formed with probability $W(U_i, U_j)$.

With this perspective in mind, we define the space of graphons.

Definition 2.2 The space $\mathcal{W}$ of **graphons** is the set of Borel measurable functions $W : [0,1]^2 \to [0,1]$ satisfying $W(x,y) = W(y,x)$, modulo the equivalence relation identifying functions that agree almost everywhere.

Remark 2.3 The almost everywhere quotient deserves comment. If W and W' differ only on a set of Lebesgue measure zero in $[0,1]^2$, then almost surely no sampled pair $\left(U_i, U_j\right)$ ever lands on the set of disagreement, and so $G(n,W)$ and $G(n,W')$ have identical distributions. Two graphons that agree almost everywhere are thus genuinely indistinguishable as random graph models, and the a.e. quotient reflects this equality rather than being a technical convenience (though this is true too).

The simplest types of graphons are multipodal graphons which are defined below.

Definition 2.4 W is a **multipodal graphon** if there exists a natural number N, a partition of $[0,1]$ by measurable sets of positive measure $I_1, ..., I_N$, and weights $w_{ij} \in [0,1]$ with $w_{ij} = w_{ji}$ for $i,j \in \{1, ..., N\}$ such that

$$W = \sum_{i=1}^{N} \sum_{j=1}^{N} w_{ij} \mathbb{1}_{I_i \times I_j}. \tag{2}$$

We call I_i the **podes** of W and the values w_{ij} the edge weights of W. When we wish to specify the number of podes, we call W an ***N*-podal graphon**. Such graphons are also called step function graphons.

Within the family of multipodal graphons lies a particularly important subclass. This subclass demonstrates exactly how graphons serve as a continuous generalization of finite graphs.

Definition 2.5 *(Graphon Blowup)* Let G be a simple graph on vertex set $\{1, ..., n\}$. Partition $[0,1]$ into intervals $I_1, ..., I_n$ of equal length, and define the graphon $W_G : [0,1]^2 \to \{0,1\}$ by

$$W_G(x,y) = \begin{cases} 1 \text{ if } (x,y) \in I_i \times I_j \text{ and } \{i,j\} \in E(G) \\ 0 \text{ otherwise} \end{cases}. \tag{3}$$

We call W_G the **graphon blowup** of G. W_G is also called the checkerboard graphon of G.

The graphon W_G is symmetric by construction and takes values in $\{0,1\}$. It encodes the adjacency matrix of G as a multipodal graphon with n equal-sized parts.

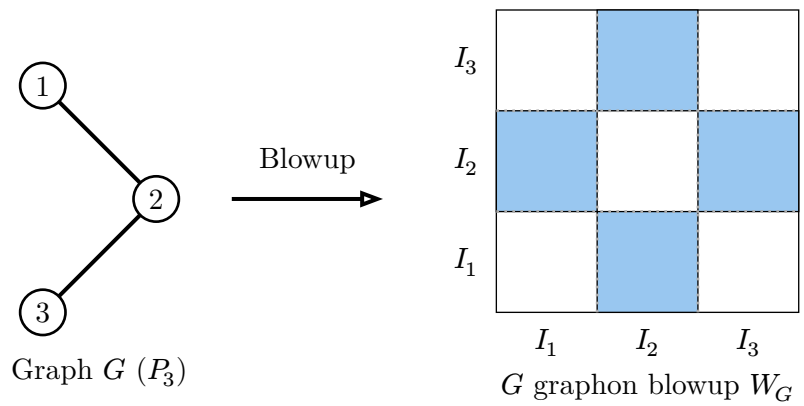

Figure 2: Graph G mapped to its graphon blowup W_G

The natural topology on $\mathcal{W}$ is induced by the **cut norm** and the induced **cut distance**.

Definition 2.6 *(Cut Norm and Cut Distance)* $\mathcal{W}$ has a natural norm, called the **cut norm** that it inherits from bounded functions from $[0,1]^2$ to $\mathbb{R}$. Where $W \in \mathcal{W}$,

$$\|W\|_\square = \sup_{S,T \in \mathcal{L}([0,1])} \left| \int_{S \times T} W(x,y)\,\mathrm{d}x\,\mathrm{d}y \right| \tag{4}$$

The cut norm induces a natural notion of distance on $\mathcal{W}$, called the **cut distance**. For $W_1, W_2 \in \mathcal{W}$,

$$d_\square(W_1, W_2) = \sup_{S,T \subset [0,1]} \left| \int_{S \times T} (W_1(x,y) - W_2(x,y))\,\mathrm{d}x\,\mathrm{d}y \right| \tag{5}$$

where $\mathcal{L}([0,1])$ is the set of Lebesgue measurable subsets of $[0,1]$.

The cut distance captures the idea that two graphons are close if, for every pair of label regions S and T, they assign the same edge mass to $S \times T$. However, $d_\square$ is too fine an equivalence for our purposes: it distinguishes graphons that ought to represent the same object. To see what additional equivalence is needed, consider the analogous issue for finite graphs.

To motivate the next quotient, recall the analogous situation for simple graphs. If G_1 and G_2 are simple graphs on the same vertex set and G_2 arises from G_1 by a permutation of vertices, that is there exists a bijection $\sigma : V(G_1) \to V(G_2)$ with $\{u,v\} \in E(G_1) \Leftrightarrow \{\sigma(u), \sigma(v)\} \in E(G_2)$, then we say G_1 is isomorphic to G_2. Any property of a graph that we care about is invariant under vertex relabeling.

The analogue for graphons is a **measure preserving bijection** $\sigma : [0,1] \to [0,1]$, which serves as a permutation of labeling space. Given such a σ and W, define $W^\sigma(x,y) = W(\sigma(x), \sigma(y))$. If $\sigma(U_1), ..., \sigma(U_n)$ are obtained from i.i.d. uniform samples $U_1, ..., U_n$, then they themselves are i.i.d. uniform on $[0,1]$. So W^σ generates the exact same distribution of random graphs as W.

Just as isomorphic simple graphs differ only in how their vertices are labeled, W and W^σ differ only in how $[0,1]$ is labeled.

Definition 2.7 *(Cut Metric)* The functional $\delta_\square$ is called the **cut metric**. For $W_1, W_2 \in \mathcal{W}$,

$$\delta_\square(W_1, W_2) = \inf_\sigma d_\square(W_1, W_2) \tag{6}$$

where the infimum is taken over the all measure preserving bijections $\sigma : [0,1] \to [0,1]$.

One of the motivations for developing the machinery of graphons, along with the cut distance and cut metric, is to establish a unified framework for comparing finite graphs, particularly those with different numbers of vertices. This comparison is achieved by embedding the discrete graphs into the continuous space $\mathcal{W}$ via their graphon blowups. In this continuous setting, the cut distance allows us to compare two graphs up to a fixed vertex labeling, whereas the cut metric inherently compares them up to graph isomorphism.

The functional $\delta_\square$ is a pseudometric on $\mathcal{W}$. It descends to a genuine metric on the quotient space defined below.

Definition 2.8 *(Reduced Graphons)*

$$\widetilde{\mathcal{W}} = \mathcal{W}/(W_1 \sim W_2 \Leftrightarrow \delta_\square(W_1, W_2) = 0) \tag{7}$$

is called the space of **reduced graphons**. The equivalence relation used to define $\widetilde{\mathcal{W}}$ is called **weak equivalence**. $\left(\widetilde{\mathcal{W}}, \delta_\square\right)$ is a metric space.

Remark 2.9 Beyond weak equivalence, another natural notion of equivalence exists. Two graphons $W_1, W_2 \in \mathcal{W}$ are considered **group equivalent** if they lie in the same orbit under the action of measure-preserving bijections on $[0,1]$. However, it is possible for the orbits of two graphons to become arbitrarily close in cut distance—yielding identical structural properties for their sampled graphs—without the graphons being strictly group equivalent. Weak equivalence resolves this issue by effectively acting as the topological closure of group equivalence.

The central topological fact, due to Lovász and Szegedy, is the following.

Theorem 2.1 *(Graphon Compactness Theorem [2])* The metric space $\left(\widetilde{\mathcal{W}}, \delta_\square\right)$ is compact.

Throughout the remainder of the thesis, we will work implicitly in the quotient space $\widetilde{\mathcal{W}}$. Furthermore, the term "graphon" will refer to the equivalence class of a reduced graphon.

To conclude these preliminaries, we return to the terminology of a "limit object." The terminology is justified by the fact that for any graphon W, the graphon blowup of $G(n, W)$ converges to W in cut metric almost surely as $n \to \infty$ [2]. Thus graphons simultaneously parametrize random graph models and arise as limits of dense graph sequences.

2.1.1. Homomorphism Densities

The primary numerical invariants of graphons are homomorphism densities, which generalize the idea of homomorphism densities of finite graphs.

Definition 2.10 Let G and H be simple graphs where $|V(G)| \leq |V(H)|$. A homomorphism from G to H is a function $f : V(G) \to V(H)$ where if $\{u, v\} \in E(G)$ then $\{f(u), f(v)\} \in E(H)$. Let $\hom(G, H)$ be the set of homomorphisms from G to H. The **homomorphism density** of G into H is

$$t(G, H) := \frac{|\hom(G, H)|}{|V(H)|^{V(G)}} \tag{8}$$

is the probability that an arbitrary function $V(G) \to V(H)$ is a homomorphism.

There is a similar notion for graphons.

Definition 2.11 For a simple graph G with k vertices and a graphon W, the **homomorphism density** of G in W is

$$t(G, W) := \int_{[0,1]^k} \prod_{\{i,j\} \in E(G)} W(x_i, x_j) \, \mathrm{d}x_1 \ldots \mathrm{d}x_k \tag{9}$$

Two special cases are central to this thesis. The **edge density** of W is

$$e(W) = \|W\|_1 = \int_{[0,1]^2} W(x, y) \, \mathrm{d}x \, \mathrm{d}y \tag{10}$$

and the **triangle density** is

$$t(W) = t(K_3, W) = \int_{[0,1]^3} W(x, y) W(y, z) W(x, z) \, \mathrm{d}x \, \mathrm{d}y \, \mathrm{d}z \tag{11}$$

From the stochastic perspective, $t(G, W)$ has a clean interpretation. It is the limiting probability, as $n \to \infty$, that a fixed map $V(G) \to V(G(n, W))$ is a graph homomorphism from G into $G(n, W)$. In particular, $e(W)$ is the edge probability and $t(W)$ is the triangle probability of the associated random graph model.

Theorem 2.2 Let G and H be simple graphs where $|V(G)| \leq |V(H)|$, then

$$t(G, H) = t(G, W_H) \tag{12}$$

where W_H is the graphon blowup of H.

A foundational property, which motivates the choice of cut topology, is that the homomorphism densities are continuous in $\widetilde{\mathcal{W}}$.

Theorem 2.3 *(Continuity of Homomorphism Density [2])* For every simple graph G, the map $t(G,\cdot)$ is continuous on $\left(\widetilde{\mathcal{W}},\delta_{\square}\right)$.

But homomorphism densities aren't just continuous.

Theorem 2.4 *(Homomorphism Density is Injective Embedding [2])* The map $\widetilde{\mathcal{W}} \to [0,1]^{\mathcal{G}}$ defined by

$$W \to (t(G,W))_{G\in\mathcal{G}} \tag{13}$$

is an injective embedding.

Together, these two theorems justify graphons as the proper limit objects for the problems we study. Because homomorphism densities, specifically the edge and triangle densities whose asymptotics we analyze, vary continuously under the cut topology, extremal questions over $\widetilde{\mathcal{W}}$ are mathematically well-posed.

2.1.2. Approximation by Finite Podal Graphons

We now return to the subclass of multipodal graphons, which will play a central role in the analysis to follow: they are often used as proxies for the full space of graphons. The following density result is what makes multipodal graphons important.

Theorem 2.5 *([2])* The set of multipodal graphons is dense in $\left(\widetilde{\mathcal{W}},\delta_{\square}\right)$. Furthermore, the set of checkerboard graphons is dense in $\left(\widetilde{\mathcal{W}},\delta_{\square}\right)$.

For multipodal graphons, the homomorphism density reduces to a finite sum of vertex to pode assignments.

Lemma 2.6 *(Reduction of Homomorphism Density)* Let W be a N-podal graphon with podes $\{S_i\}_{i=1}^N$ and edge weights w_{ij}. For any simple graph G,

$$t(G,W) = \sum_{\varphi:V(G)\to\{1,\dots,N\}} \left(\prod_{v\in V(G)} \mu\left(S_{\varphi(v)}\right)\right)\left(\prod_{\{i,j\}\in E(G)} w_{\varphi(i)\varphi(j)}\right) \tag{14}$$

Proof Let $V(G)=\{1,\dots,k\}$. Since $\{S_i\}_{i=1}^N$ partitions $[0,1]$, the product space $[0,1]^k$ is partitioned into cells $\prod_{i=1}^k S_{\varphi(i)}$ indexed by functions $\varphi : V(G)\to\{1,\dots,N\}$. Hence

$$\begin{aligned} t(G,W) &= \int_{[0,1]^k} \prod_{\{i,j\}\in E(G)} W\left(x_i,x_j\right) \mathrm{d}x_1 \dots \mathrm{d}x_k \\ &= \sum_{\varphi:V(G)\to\{1,\dots,N\}} \int_{\prod_{a=1}^k S_{\varphi(a)}} \prod_{\{i,j\}\in E(G)} W\left(x_i,x_j\right) \mathrm{d}x_1 \dots \mathrm{d}x_k. \end{aligned} \tag{15}$$

By the definition of W,

$$
\begin{aligned}
t(G,W) &= \sum_{\varphi:V(G)\to\{1,\dots,N\}} \int_{\prod_{a=1}^{k} S_{\varphi(a)}} \prod_{\{i,j\}\in E(G)} w_{\varphi(i)\varphi(j)}\,\mathrm{d}x_1\dots\mathrm{d}x_k \\
&= \sum_{\varphi:V(G)\to\{1,\dots,N\}} \left(\prod_{\{i,j\}\in E(G)} w_{\varphi(i)\varphi(j)}\right) \int_{\prod_{a=1}^{k} S_{\varphi(a)}} \mathrm{d}x_1\dots\mathrm{d}x_k \\
&= \sum_{\varphi:V(G)\to\{1,\dots,N\}} \left(\prod_{\{i,j\}\in E(G)} w_{\varphi(i)\varphi(j)}\right)\left(\prod_{v\in V(G)} \mu\big(S_{\varphi(v)}\big)\right)
\end{aligned}
\tag{16}
$$

□

Specializing the reduction lemma to $G = K_2$ and $G = K_3$, for which $\varphi : V(G) \to \{1, ..., N\}$ are ordered pairs and triples respectively, gives the following expressions for the edge and triangle densities

$$
\begin{aligned}
e(W) &= \sum_{i,j=1}^{N} \mu(S_i)\mu\big(S_j\big)w_{ij} \\
t(W) &= \sum_{i,j,k=1}^{N} \mu(S_i)\mu\big(S_j\big)\mu(S_k)w_{ij}w_{ik}w_{jk}
\end{aligned}
\tag{17}
$$

2.1.3. Razborov Triangle

The image of the function $(e,t) : \widetilde{\mathcal{W}} \to [0,1]^2$ where $W \to (e(W), t(W))$ is called the **Razborov Triangle $\mathcal{R}$**, after Razborov found the minimal triangle density for a given edge density in his seminal paper [8].

Theorem 2.7 *(Maximum Triangle Density)* For a graphon W with edge density ε

$$
t(W) \le \varepsilon^{\frac{3}{2}}. \tag{18}
$$

The bound is uniquely satisfied by $W = \mathbb{1}_{[0,\sqrt{\varepsilon}]^2}$.

Theorem 2.8 *(Minimum Triangle Density I (Mantel's Theorem))* For a graphon W with edge density $\varepsilon \in \left[0, \frac{1}{2}\right]$, $t(W) \ge 0$. An example of a graphon which achieves this lower bound is

$$
W = 2\varepsilon\Big(\mathbb{1}_{[0,\frac{1}{2}]\times[\frac{1}{2},1]} + \mathbb{1}_{[\frac{1}{2},1]\times[0,\frac{1}{2}]}\Big) \tag{19}
$$

Theorem 2.9 *(Minimum Triangle Density II [8])* For a graphon W with edge density $\varepsilon \in \left[1-\frac{1}{k}, 1-\frac{1}{k+1}\right]$ for $k \ge 2$

$$
t(W) \ge \frac{k(k-1)}{(k+1)^2}\left(1+\sqrt{1-\varepsilon\frac{k+1}{k}}\right)^2\left(1-2\sqrt{1-\varepsilon\frac{k+1}{k}}\right) \tag{20}
$$

Remark 2.12 Razborov proved this using Flag Algebras, which can be thought of as a dual object to graphons by Theorem 2.4. Flag Algebras seek to understand how the collections of homomorphism densities behave. Furthermore, one can define an algebra over such objects to do interesting computations, which often reduce to semidefinite programming problems. See [8], [12], [13] for further reference on this topic.

Theorem 2.10 *(Minimum Triangle Density III [14])* Let $\varepsilon \in \left[1 - \frac{1}{k}, 1 - \frac{1}{k+1}\right]$. Let

$$c = \frac{1 + \sqrt{1 - \varepsilon \frac{k+1}{k}}}{k+1}. \tag{21}$$

Partition $[0, 1]$ into k intervals $I_1, ..., I_k$ where $|I_i| = c$ for $i \in \{1, ..., k-1\}$ and $|I_k| = 1 - (k-1)c$. Define a graphon W as follows. We set $W = 1$ when $I_i \times I_j$ for all $i \neq j$. For $i < k$, $W = 0$ on $I_i \times I_i$. Finally, on $I_k \times I_k$ the graphon is not constrained, except to have the correct average value, and to not allow any triangles where all three vertices are in I_k. For such a graphon $t(W)$ achieves the lower bound in Theorem 2.9.

Remark 2.13 This strategy, partitioning the graphon such that one localized region exhausts the allowed triangle density while the remainder contributes to the edge density without forming triangles, will be directly reflected in the constructions later in this thesis.

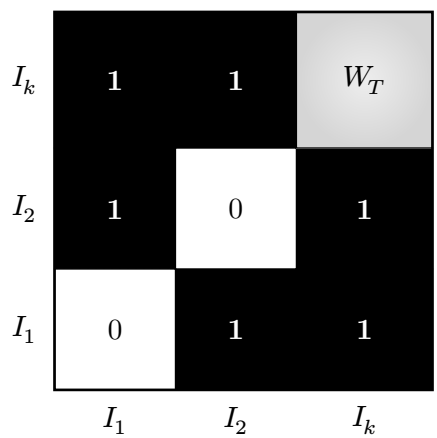

Figure 3: Extremizing graphon for $\varepsilon \in \left[1 - \frac{1}{3}, 1 - \frac{1}{4}\right]$

Putting the results of Theorem 2.7, Theorem 2.8, Theorem 2.9, and Theorem 2.10 together, we see $\mathcal{R}$ looks like the region below in Figure 4.

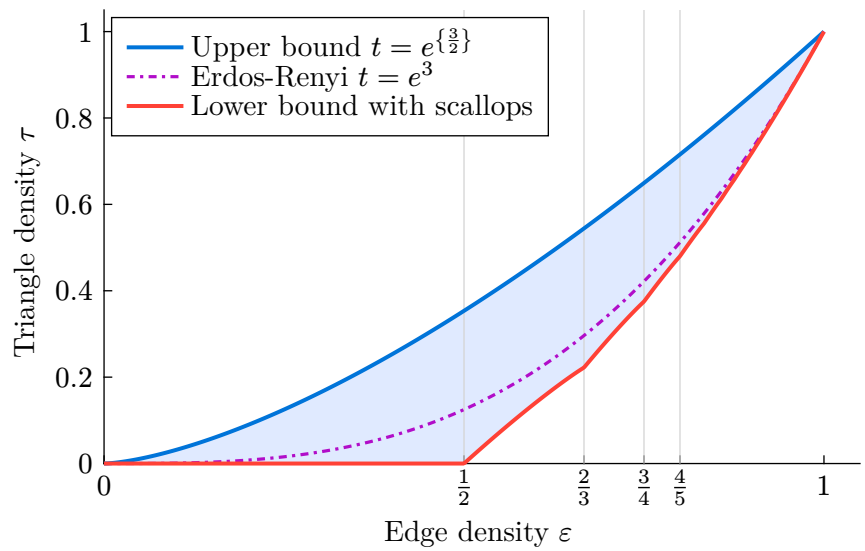

Figure 4: The Razborov Triangle $\mathcal{R}$: The shaded interior represents the achievable triangle densities t for a given edge density e.

Our region of interest in this thesis is looking infinitesimally close to $(0, 0.5] \times \{0\} \in R$.

2.2. Linear Programming

Lets switch gears a bit, and discuss Linear Programming. We will follow the notation and terminology of Vanderbei, R.J. [10]. A **Linear Program (LP)** is a optimization problem in which a linear function is minimized or maximized subject to finitely many linear inequality or equality constraints. The general minimization form which we will use is

$$\min c^T x \text{ subject to } Ax \geq b, x \geq 0, \tag{22}$$

where $A \in \mathbb{R}^{m \times n}$, $c \in \mathbb{R}^n$, $b \in \mathbb{R}^m$ are the **input data** and $x \in \mathbb{R}^n$ is the vector of **variables**. The linear function is called the **objective function**, and the conditions $Ax \geq b$ and $x \geq 0$ (both interpreted component wise) are the **constraints**.

A vector $x \in \mathbb{R}^n$ which satisfies all the constraints is a **feasible point**; the set of all feasible points is the **feasible region**, which is a polyhedron, an intersection of a collection of half spaces. An LP is **feasible** if the feasible region is nonempty, and bounded if $c^T x$ is bounded below on the feasible region. The **optimal value** is the infimum $\boldsymbol{c^T x}$ over the feasible region, and a feasible point achieving this value is an **optimal solution**.

Other standard forms (maximization, $\leq$, equality constraints, or free variables) can be converted into the form above by standard transformations, and all results we state below apply equally in those forms.

Geometrically, an LP asks for the minimum of a linear function over a polyhedron. Since the objective is linear, whenever an optimal solution exists, one can be found at a vertex of the feasible region. This underlies much of the theory.

Example 2.14 Consider the LP $\min x + y$ subject to the constraints that

$$\begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} \geq \begin{pmatrix} 2 \\ 2 \end{pmatrix} \tag{23}$$

and $x, y \geq 0$. The feasible region is an unbounded polyhedron in the first quadrant lying above the lines $x + 2y = 2$ and $2x + y = 2$. The level sets of the objective function are lines of slope -1. Thus starting at $x + y = 0$ and then moving the line upward, the first point we hit of the feasible region is the intersection of the two boundary lines, namely $(x, y) = \left(\frac{2}{3}, \frac{2}{3}\right)$, with objective value $\frac{4}{3}$.

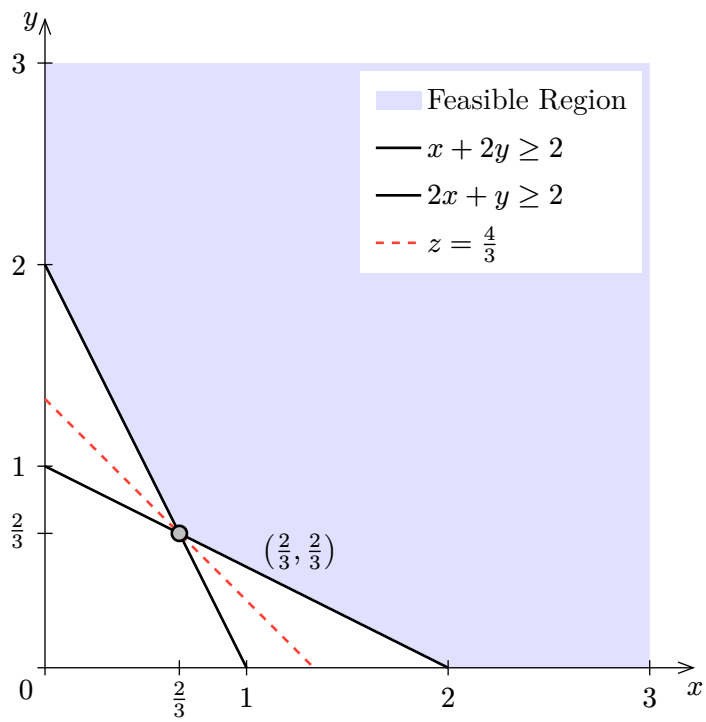

Figure 5: Graphical solution to the linear program.

Two features seen in the example will recur. First the optimum is obtained at a vertex of the feasible region. It will specifically occur at the intersection of two binding constraints. Second, although the input data are integers, the optimal solution has rational (non-integer) coordinates; this is typical of LPs arising from combinatorial problems and reflects the fact that an LP is often a relaxation of a discrete problem.

2.2.1. Existence of Solutions to an LP

Given an LP of the form above. We have 3 possibilities regarding its behavior:

1. The LP is **infeasible**: There exists no $x \in \mathbb{R}^n$ satisfying the constraints
2. The LP is **feasible** but **unbounded**: For all $M \in \mathbb{R}$, there exists $x \in \mathbb{R}^n_{\geq 0}$ such that $c^T x < M$.
3. The LP is **feasible and bounded**: $c^T x$ has a finite infimum over the feasible region

In the third case, it is not obvious a priori that the infimum is attained. The feasible region is closed (as an intersection of finitely many closed half-spaces) but it need not be compact (see Example 2.14), and a continuous function on a non-compact closed set need not attain its infimum. However a foundational result in the theory of LPs tells us that the infimum is always attained.

Theorem 2.11 *(Existence of Optimal Solutions)* If an LP is feasible and bounded, then it has an optimal solution; moreover, one can always be chosen at a vertex of the feasible region

The second part of the theorem gives the intuition on why the first part holds. Although the polyhedron is unbounded, its set of vertices form a finite set, and a minimum of a linear function over the feasible region, if finite, must be attained at one of them. A full proof uses the structure of the polyhedra (every feasible point is a convex combination of the vertices plus a nonnegative combination of extreme rays) and the observation that the objective can only decrease by moving along the extreme ray if the LP is unbounded.

The LPs that occur in the thesis are of a particularly nice form. Their feasible regions are nonempty. Every LP we will consider has a feasible point obtained by setting all the variables sufficiently large. Furthermore, their objectives are bounded below by 0, since all the variables are nonnegative and the objective coefficients are nonnegative. Hence every LP we consider has an optimal solution.

2.2.2. Dual Linear Programs

For linear programs, there is a natural notion of duality, which we will define below. Dual Linear Programs and the theorems relating a LP and its dual, will be a vital tool in this thesis.

Definition 2.15 Given the **primal** LP,

$$\min c^T x \text{ subject to } Ax \geq b, x \geq 0, \tag{24}$$

its **dual** is the LP

$$\max b^T y \text{ subject to } A^T y \leq c, y \geq 0, \tag{25}$$

Variables of the dual are called **dual variables** and constraints of the dual are called **dual constraints**. By construction there is one dual variable per primal constraint, and one dual constraint per primal variable.

The dual of the dual, formed by applying this transformation again (after converting the maximization to standard minimization form), is the primal. So **primal** and **dual** are symmetric. The argument above gives us the following theorem for free.

Theorem 2.12 *(Weak Duality)* Let x be a feasible point of the primal LP and y be a feasible point of the dual LP. Then

$$c^T x \geq b^T y. \tag{26}$$

In particular, the optimal value of the dual is at most the optimal value of the primal.

Proof $b^T y = y^T b \leq y^T Ax = \left(A^T y\right)^T x \leq c^T x$, where we are using the constraints of x and y. $\square$

Weak duality tells us the dual gives a lower bound on the primal. The fundamental theorem of LP duality says that this bound is in fact tight.

Theorem 2.13 *(Strong Duality)* Suppose the primal and dual are both feasible. Then they have optimal solutions, and their optimal values are equal:

$$\min\{c^T x : Ax \geq b, x \geq 0\} = \max\{b^T y : A^T y \leq c, y \geq 0\} \tag{27}$$

Strong Duality is a deep result. Its proof relies on a classical theorem of linear inequalities called Farkas' lemma. We refer to Vanderbei, R.J. [10] for full treatment. For our purposes, Strong Duality will be the key structural tool in the thesis. It allows us to prove statements about the primal optimum by reasoning about the dual, and vice versa.

Linear programs of the form considered here, minimizing a linear functional over a polyhedron defined by combinatorially-indexed constraints, arise throughout combinatorics as the natural continuous analogues to discrete optimization problems over graphs. LP duality then provides a structural link between the problem and its dual that is often surprising and powerful. In Section 5 we will define a linear program $\alpha(G)$, whose constraints are indexed by the triangles of a graph G. Strong duality applied to $\alpha(G)$ will be essential to the upper bound arguments of later sections.

2.3. Finner's Generalized Holder Inequality

The following is a classical inequality in measure theory, together with a corollary that will be convenient for our application.

Theorem 2.14 *(Finner's Generalized Hölder Inequality [9])* Let $n \in \mathbb{N}$, and let $\{(X_i, \mathcal{M}_i, \mu_i)\}_{i=1}^n$ be a collection of measure spaces. Let $X = X_1 \times ... \times X_n$ be their full product, equipped with the product measure μ. For any nonempty subset of indices $S \subset [n]$, let $X_S = \prod_S X_k$ denote the lower-dimensional product space with the corresponding product measure μ_S. Suppose we have m non-empty index subsets $S_1, ..., S_m \subset [n]$, and corresponding exponents $p_1, ..., p_m \in [1, \infty)$. Assume that for every coordinate $k \in \{1, ..., n\}$, the sum of $\frac{1}{p_j}$ over the subsets S_j that contain k is exactly 1:

$$\sum_{\{j : k \in S_j\}} \frac{1}{p_j} = 1 \tag{28}$$

For each $j \in [m]$, let $f_j : X_{S_j} \to \mathbb{R}$ be a measurable function such that $f_j \in L^{p_j}\left(X_{S_j}, \mu_{S_j}\right)$. Then, the product $\prod_{i=1}^m |f_i|$, evaluated as a function on the full space X, is integrable with respect to μ, and the following inequality holds:

$$\int_X \prod_{i=1}^{m} \left| f_i\left(x_{S_i}\right) \right| \mathrm{d}\mu \leq \prod_{i=1}^{m} \|f_i\|_{p_i} \tag{29}$$

Corollary 2.15 Let $n \in \mathbb{N}$, and let $\{(X_i, \mathcal{M}_i, \mu_i)\}_{i=1}^{n}$ be a collection of probability spaces. Let $X = X_1 \times ... \times X_n$ be their full product, equipped with the product measure μ. For any nonempty subset of indices $S \subset [n]$, let $X_S = \prod_S X_k$ denote the lower-dimensional product space with the corresponding product measure μ_S. Suppose we have m nonempty index subsets $S_1, ..., S_m \subset [n]$, and corresponding exponents $p_1, ..., p_m \in [1, \infty)$. Assume that for every coordinate $k \in \{1, ..., n\}$, the sum of $\frac{1}{p_j}$ over the subsets S_j that contain k is less than or equal to 1:

$$\sum_{\{j:k \in S_j\}} \frac{1}{p_j} \leq 1 \tag{30}$$

For each $j \in [m]$, let $f_j : X_{S_j} \to \mathbb{R}$ be a measurable function such that $f_j \in L^{p_j}\left(X_{S_j}, \mu_{S_j}\right)$. Then, the product $\prod_{i=1}^{m} |f_i|$, evaluated as a function on the full space X, is integrable with respect to μ, and the following inequality holds:

$$\int_X \prod_{i=1}^{m} \left| f_i\left(x_{S_i}\right) \right| \mathrm{d}\mu \leq \prod_{i=1}^{m} \|f_i\|_{p_i} \tag{31}$$

Proof Let $A = \{a_1, ..., a_k\}$ be the set of coordinates where

$$\sum_{\{j:a_i \in S_j\}} \frac{1}{p_j} < 1. \tag{32}$$

Let $S_{m+1} = \{a_1\}, ..., S_{m+k} = \{a_k\}$. Let

$$p_{m+i}^{-1} = 1 - \sum_{\{j:a_i \in S_j, j \leq m\}} p_j^{-1}, \tag{33}$$

and let $f_{m+i} = 1$ for all $i \in [k]$. Thus

$$\begin{aligned} \int_X \prod_{i=1}^{m} \left| f_i\left(x_{S_i}\right) \right| \mathrm{d}\mu &= \int_X \prod_{i=1}^{m+k} \left| f_i\left(x_{S_i}\right) \right| \mathrm{d}\mu \\ &\leq \prod_{i=1}^{m+k} \|f_i\|_{p_i} = \prod_{i=1}^{m} \|f_i\|_{p_i}, \end{aligned} \tag{34}$$

where the first equality comes from the fact that $f_{S_{m+i}} \equiv 1$. The inequality comes from Theorem 2.14. The last inequality comes from the fact $\left\|f_{m+i}\right\|_{p_{m+i}} = \mu\left(X_{a_i}\right)^{\frac{1}{p_{a_i}}}$ (using the probability space assumption). □

2.4. Notation

Sets: For any natural number N, define $[N] := \{1, ..., N\}$. For a finite set S and a natural number $n < |S|$, we write $\binom{S}{n}$ for the set of all n-element subsets of S. Let $\mathbb{R}_{\geq 0} := [0, \infty)$, $\mathbb{R}_{>0} := (0, \infty)$. Let μ be the Lebesgue measure on $[0, 1]$ or on $[0, 1]^2$, where context will make it clear whether we are talking about $[0, 1]$ or $[0, 1]^2$.

Graphs: For a graph G, we will write $V(G)$ and $E(G)$ as the vertex and edge sets, respectively. An edge between vertices u and v is written as $\{u, v\}$ or sometimes uv when there is no ambiguity. For a subgraph $T \subset G$, we write $V(T)$ and $E(T)$ for its vertex and edge sets. We write $\mathcal{T}(G)$ for the set of triangles in G. A triangle in G is a 3-vertex subgraph of G isomorphic to K_3. For any natural number n, let K_n be a complete graph on n vertices, in other words $K_n := \left([n], \binom{[n]}{2}\right)$ as a vertex set and edge set pair. In particular, K_2 is an edge and K_3 is a triangle.

Graphons: For (ε, τ) in the Razborov Triangle, we will define the following spaces of graphons for convenience:

$$\begin{aligned} \widetilde{\mathcal{W}}(N) &:= \left\{W \;\middle|\; W \in \widetilde{\mathcal{W}}, W \text{ is } N\text{-podal}\right\} \\ \widetilde{\mathcal{W}}(\varepsilon, \tau) &:= \left\{W \;\middle|\; W \in \widetilde{\mathcal{W}}, e(W) = \varepsilon, t(W) \leq \tau\right\} \\ \widetilde{\mathcal{W}}(N; \varepsilon, \tau) &:= \widetilde{\mathcal{W}}(N) \cap \widetilde{\mathcal{W}}(\varepsilon, \tau). \end{aligned} \tag{35}$$

Note if (ε, τ) are not in the Razborov Triangle, $\widetilde{\mathcal{W}}(\varepsilon, \tau)$ will be empty, by definition.

Asymptotics: For functions $f, g : D \to \mathbb{R}_{>0}$ and a limit point $p \in D$, we write $f(x) = O(g(x))$ if there exists $C > 0$ such that $f(x) \leq Cg(x)$ in a punctured neighborhood of p. We write $f(x) = \Omega(g(x))$ if there exists $c > 0$ such that $f(x) \geq cg(x)$ in a punctured neighborhood of p. We write $f(x) = \Theta(g(x))$ if $f(x) = O(g(x))$ and $f(x) = \Omega(g(x))$.

3. The Extremal Problem

Having developed the relavent machinery, we will state the problem this thesis addresses. Fix a simple graph G, and a pair of edge and triangle densities in the Razborov triangle $(\varepsilon, \tau) \in \operatorname{int}(\mathcal{R})$. We ask: among all graphons with edge density ε and triangle density at most τ, what is the tightest upper bound we can get for the homomorphism density of G? That is, we study the quantity

$$T^G_{\max}(\varepsilon, \tau) := \sup\left\{t(G, W) \;\middle|\; W \in \widetilde{\mathcal{W}}(\varepsilon, \tau)\right\} \tag{36}$$

and how it behaves near the boundary of the Razborov Triangle. In this thesis, we will seek to understand the asymptotics of $T^G_{\max}$ for $\tau \to 0$ and $\varepsilon \in \left(0, \frac{1}{2}\right]$.

Intuitively, this is an extremal question about how much subgraph structure G can survive in a graphon with a few triangles. The regime $\tau \to 0$ forces the graphon toward triangle-free behavior while holding the edge density, and the question is whether (and how quickly) G-density must also decay.

Before turning to the asymptotics, we note that $T^G_{\max}(\varepsilon, \tau)$ is a genuine maximum, and not just a supremum.

Proposition 3.1 For every simple graph G, every $(\varepsilon, \tau) \in \mathcal{R}$, there exists a graphon $W \in \widetilde{\mathcal{W}}(\varepsilon, \tau)$ such that $t(G, W) = T^G_{\max}(\varepsilon, \tau)$.

Proof Let $F : \widetilde{\mathcal{W}} \to \mathcal{R} \subset [0,1]^2$ where $F(W) = (e(W), t(W))$. By Theorem 2.3, e and t are continuous, meaning F is continuous. By definition, $\widetilde{\mathcal{W}}(\varepsilon, \tau) = F^{-1}(\varepsilon \times [0, \tau])$. $\varepsilon \times [0, \tau]$ is closed, and as the preimage of a closed set $\widetilde{\mathcal{W}}(\varepsilon, \tau)$ is closed. Since $\left(\widetilde{\mathcal{W}}, \delta_\square\right)$ is compact and all closed sets of a compact set are compact, $\widetilde{\mathcal{W}}(\varepsilon, \tau)$ is compact. Finally, the map $t(G, \cdot)$ is continuous by Theorem 2.3, and a continuous real-valued function on a nonempty compact set attains its supremum. Therefore, $T^G_{\max}(\varepsilon, \tau)$ is attained by some $W \in \widetilde{\mathcal{W}}(\varepsilon, \tau)$. $\square$

So we can redefine $T^G_{\max}(\varepsilon, \tau)$ as

$$T^G_{\max}(\varepsilon, \tau) := \max\Big\{ t(G, W) \;\Big|\; W \in \widetilde{\mathcal{W}}(\varepsilon, \tau) \Big\}. \tag{37}$$

To show $T^G_{\max}(\varepsilon, \tau) = \Theta(g(\tau))$ for some function g, we have to show both $O(g(\tau))$ and $\Omega(g(\tau))$. $\Omega(g(\tau))$ is a bit finicky but straight forward, for fixed $\varepsilon \in (0, 0.5]$ we just have to exhibit a continuous function $W : (0, \tau_*] \to \widetilde{\mathcal{W}}$ for some $\tau_* > 0$ where $\tau \to W_\tau \in \widetilde{\mathcal{W}}(\varepsilon, \tau)$. We show that $t(G, W_\tau) = g(\tau)$ as $\tau \to 0$. We do this in Proposition 7.2 and Proposition 7.3 for $\varepsilon \in (0, 0.5)$ and $\varepsilon = 0.5$ respectively. $O(g(\tau))$ requires a bit more work, which is why we look for simplifications to give us intuition, and hopefully reductions of the problem.

4. Triangle Spanning Decomposition

To establish the upper bound,we first prove the result for a simpler family of graphs: those where every edge belongs to a triangle (see Proposition 7.1). In this section, we demonstrate that any arbitrary graph can be decomposed into a set of these triangle-spanning subgraphs, denoted $\mathcal{C}$, alongside a collection of isolated edges and vertices. Subsequently, in Section 7, we will show that these residual components do not asymptotically affect the homomorphism density, allowing us to cleanly reduce the general upper bound to the triangle-spanning case.

Definition 4.1 A simple graph G is triangle spanning if every edge of G lies in a triangle of G.

Definition 4.2 Let G be a simple graph. For triangles $S, T \in \mathcal{T}(G)$, $S \leftrightarrow T$ if theres a sequence $S = T_0, T_1, ..., T_k = T$ of triangles in G where $V(T_i) \cap V(T_{i+1}) \neq \emptyset$ for all i. Let $\mathcal{T}_1, ..., \mathcal{T}_n$ be the equivalence classes under $\leftrightarrow$. Define C_i a subgraph of G with vertex set

$$V(C_i) := \bigcup_{T \in \mathcal{T}_i} V(T) \tag{38}$$

and edge set

$$E(C_i) := \bigcup_{T \in \mathcal{T}_i} E(T). \tag{39}$$

Define the **bridge edges** $B := E(G) \setminus \bigcup_i E(C_i)$, and the **leftover vertices** $L := V(G) \setminus \bigcup_i V(C_i)$.

Lemma 4.1 The C_i are vertex disjoint and edge disjoint triangle spanning subgraphs of G, B consists of edges not lying in any triangle of G, and L be the vertices in no triangles, $\{E(C_i)\}_{i=1}^n \cup B$ is a partition of $E(G)$, and $\{V(C_i)\}_{i=1}^n \cup L$ is a partition of $V(G)$.

Proof

1. **B consists of edges not in any triangle of G**: Let $e \in E(G)$. If e lies in some triangle $T \in \mathcal{T}(G)$, then $T \in \mathcal{T}_i$ for some i, so $e \in E(T) \subset E(C_i)$. Hence $e \notin B$. Conversely, if $e \in C_i$, then by definition $e \in E(T)$ for some triangle $T \in \mathcal{T}_i \subset \mathcal{T}(G)$, so e lies in some triangle of G. Thus $e \in B$ if and only if e lies in no triangle of G. The same argument with vertices in place of edges shows $v \in L$ if and only if v lies in no triangle.
2. **$\{E(C_i)\}$ and $\{V(C_i)\}$ are pairwise disjoint**: Suppose there exists $e \in C_i \cap C_j$ for some $i, j \in [n]$. Then $e \in E(A)$ for some $A \in \mathcal{T}_i$ and $B \in \mathcal{T}_j$. Since A and B share an edge e, hence share two vertices, so $A \leftrightarrow B$ directly (chain of length 1). So $i = j$. There for $E(C_i)$ are pairwise disjoint. The same argument works for the vertices. Combining with the definitions for B and L, this gives us the partitions $E(G) = \sqcup_i E(C_i) \sqcup B$ and $V(G) = \sqcup_i V(C_i) \cup L$.
3. **Each C_i is triangle spanning**: By construction, $\forall e \in E(C_i)$ there exists $T \in \mathcal{T}_i$ such that $e \in E(T)$. We must show T is a triangle in C_i, but this is trivial. By construction, $V(T) \subset V(C_i)$ and $E(T) \subset E(C_i)$. So e is contained in a triangle T in C_i.

□

Definition 4.3 The C_i are the **maximal triangle-spanning components** of G. The triple $(\mathcal{C}, B, L)$, where $\mathcal{C} = \{C_i\}_{i=1}^n$, is the **triangle spanning decomposition** of G. Let $V(\mathcal{C}) := V(G) \setminus L$ and $E(\mathcal{C}) := E(G) \setminus B$.

Thus we will prove the desired upper bound for triangle spanning graphs first, then we will use the decomposition and reduce the general graph case to a finite number of triangle spanning subgraphs.

5. The Graph Parameter $\alpha(G)$

In this section, we will introduce the graph parameter $\alpha(G)$ that will control the asymptotics of $T^G_{\max}(\varepsilon, \tau)$. Before the formal definition, we will motivate it through a heuristic computation.

Suppose, for a moment, that $T^G_{\max}(\varepsilon, \tau) = \Theta(\tau^\alpha)$ for some $\alpha = \alpha(G) \geq 0$, possibly depending on G.

This assumption is a guess. A priori, $T^G_{\max}$ could decay faster than any power of τ, slower than any power, or may not have any clean scaling. But the guess is motivated by the fact the relavent functionals are all polynomials: on a multipodal graphon with pode measure α_i and edge weights w_{ij}, both $t(W)$ and $t(G, W)$ are polynomial in α_i and edge weights w_{ij}. So a question about one polynomial functional constrained by other has the chance of admitting a clean power-law scaling. Not automatic, but plausible enough to pursue.

Taking logarithms and passing to the limit we see the exponent would be

$$\alpha = \lim_{\tau \to 0} \frac{\log\big(T^G_{\max}(\varepsilon, \tau)\big)}{\log(\tau)}, \tag{40}$$

if the limit exists.

Since $T^G_{\max}$ is the maxiumum of $t(G, W)$ over $\widetilde{\mathcal{W}}(\varepsilon, \tau)$, we can approximate α by exhibiting a graphon $W_\tau \in \widetilde{\mathcal{W}}(\varepsilon, \tau)$ which depends on τ whose $t(G, W_\tau)$ exhibits a clean scaling. Any such construction gives a bound on the exponent; the right construction should give its value.

Candidate Construction: Assume G is a triangle spanning graph and has a triangle. We can technically do this for all G, but the triangle spanning case is a lot more illuminating. Our guiding intuition is to avoid "wasted" triangles: any triangle formed in our graphon should ideally correspond to a triangle actually present in G. the graphon generates extraneous triangles, it exhausts our tightly constrained τ-budget without increasing the density of G.

To realize this, consider constructing a multipodal graphon with $V = |V(G)|$ podes, where each pode corresponds to a vertex of G. We scale the sizes of these podes and the weights of the edges between them as powers of τ. Let the measure of the v-th pode be τ^{x_v} for some $x_v \geq 0$, let the edge weight between pode u and v be roughly $\tau^{y_{uv}}$ (for non edges in G, set the edge weight to 0).

With this construction, the homomorphism density of G generated by mapping each vertex of G to its corresponding pode is proportional to the product product of the pode measures and edge weights:

$$t(G, W) \approx \prod_{v \in V(G)} \tau^{x_v} \prod_{e \in E(G)} \tau^{y_e} = \tau^{\sum_{v \in V(G)} x_v + \sum_{e \in E(G)} y_e} \tag{41}$$

Because τ is small ($\tau \to 0$), to *maximize* this density, we must minimize the exponent $\sum_{v \in V(G)} x_v + \sum_{e \in E(G)} y_e$. However, we have a restricted total triangle budget. In this V-podal graphon, the triangles formed correspond exactly to $T \in \mathcal{T}(G)$. The density of any specific triangle T in the graphon scales as:

$$\tau^{\sum_{v\in V(T)} x_v^* + \sum_{e\in E(T)} y_e^*}. \tag{42}$$

To satisfy the constraint that $t(W) \leq \tau = \tau^1$, every individual triangle formed in the graphon must have a density of $O(\tau)$. Therefore, the exponent for each triangle must be at least 1. While this ansatz is not wholly correct, one can notice that $e(W) \to 0$ as $\tau \to 0$. But there is quite an easy fix to this situation, inspired by Theorem 2.10, by adding some podes which soak up the remaining edge density while creating no triangles.

This heuristic directly translates our structural embedding problem into a Linear Program. To find the optimal power-law exponent for $T^G_{\max}$, we must minimize the exponent of the G-density subject to the constraint that no triangle exceeds the τ^1 budget. This optimal exponent is exactly the graph parameter $\alpha(G)$.

Definition 5.1 Let $\alpha : \mathcal{G} \to \mathbb{R}_{>0}$ where

$$\alpha(G) := \min\left\{ \sum_{v\in V(G)} x_v + \sum_{e\in E} y_e \right\} \tag{43}$$

where $x : V(G) \to \mathbb{R}_{\geq 0}$, $y : E(G) \to \mathbb{R}_{\geq 0}$ and for all triangles τ in G,

$$\sum_{v\in V(\tau)} x_v + \sum_{e\in E(\tau)} y_e \geq 1. \tag{44}$$

The heuristic above motivates $\alpha(G)$ as the smallest exponent attainable by a construction of this type; it is not a priori clear that no other construction achieves a smaller exponent, or indeed that the scaling $\Theta\big(\tau^{\alpha(G)}\big)$ is achieved at all. Establishing both, that $\alpha(G)$ is the exponent, not merely a candidate, is the the content of the main theorem of this thesis, proved in the sections that follow.

Definition 5.2 The dual LP to $\alpha(G)$ is defined as follows:

$$\alpha^*(G) := \max\left\{ \sum_{T\in\mathcal{T}(G)} z_T \right\} \tag{45}$$

where $z : \mathcal{T}(G) \to \mathbb{R}_{\geq 0}$ and for all vertices $v \in V(G)$ and for all $e \in E(G)$

$$\begin{aligned} \sum_{\substack{T\in\mathcal{T}(G)\\ T\ni v}} z_T &\leq 1 \\ \sum_{\substack{T\in\mathcal{T}(G)\\ T\ni e}} z_T &\leq 1 \end{aligned} \tag{46}$$

With the primal and dual linear programs established, we prove a brief but essential structural property of the optimal primal solution (x^*, y^*). In our lower bound, constructions later in the thesis, we will build graphons parameterized exactly by these optimal LP variables. To ensure our constructed graphon utilizes the full triangle budget, meaning it achieves a triangle density of exactly $\Theta(\tau)$ rather than something asymptotically smaller, we must guarentee that the primal optimal solution doesn't loosely over-satisfy all of its constraints. The following lemma guarentees that at least one triangle constraint is strictly tight, effectively anchoring our construction to the τ boundary.

Lemma 5.1 Let G be a simple graph where $\mathcal{T}(G) \neq \emptyset$. Let x^* and y^* be the vertex and edge weight functions which are the solutions to $\alpha(G)$. There exists a $T \in \mathcal{T}(G)$ such that

$$\sum_{v \in V(T)} x_v^* + \sum_{e \in E(T)} y_e^* = 1 \tag{47}$$

Proof Assume the contrary that for all $T \in \mathcal{T}(G)$

$$\sum_{v \in V(T)} x_v^* + \sum_{e \in E(T)} y_e^* > 1. \tag{48}$$

Thus there exists a $\delta > 0$ such that for all $T \in \mathcal{T}(G)$

$$\sum_{v \in V(T)} x_v^* + \sum_{e \in E(T)} y_e^* \geq 1 + \delta. \tag{49}$$

Fix some $T_0 \in \mathcal{T}(G)$. Let $S = \{x_v^* \mid v \in V(T_0)\} \cup \{y_e^* \mid e \in E(T_0)\}$. Let $z = \max S$ and f be the object which gives z (ie if f is a vertex $z = x_f$ and if its an edge $z = y_f$).

$$(|V(G)| + |E(G)|)z \geq \sum_{v \in V(T_0)} x_v^* + \sum_{e \in E(T_0)} y_e^* \geq 1 + \delta. \tag{50}$$

Thus

$$z \geq \frac{1 + \delta}{|V(G)| + |E(G)|} > 0 \tag{51}$$

Let $\delta' = \min(z, \delta)$. Let $(\tilde{x}, \tilde{y})$ be identical to (x^*, y^*) except at f where $\tilde{z}_f = z_f - \delta'$. The only sums which change are the sums which f is a part of. For all $T \in \mathcal{T}(G)$ such that $f \in V(T)$ or $f \in E(T)$,

$$\begin{aligned} \sum_{i \in V(T)} \tilde{x_i} + \sum_{e \in E(T)} y_e^* &= \sum_{i \in V(T)} x_i^* + \sum_{e \in E(T)} y_e^* - \delta' \\ &\geq 1 + \delta - \delta' \geq 1 \end{aligned} \tag{52}$$

because $\delta \geq \delta'$ by definition. Finally,

$$\begin{aligned}\sum_{v\in V(G)} \tilde{x}_v + \sum_{e\in E(G)} y_e^* &= \sum_{v\in V(G)} x_v^* + \sum_{e\in E(G)} y_e^* - \delta' \\ &= \alpha(G) - \delta' < \alpha(G)\end{aligned} \tag{53}$$

contradicting optimality. □

We conclude this section by bridging our algebraic parameter with our graph-theoretic decomposition. Because the bridge edges B and leftover vertices L of our triangle-spanning decomposition do not participate in any triangles of G, their corresponding variables x_v and y_e do not appear in any constraints of the primal LP for $\alpha(G)$

Proposition 5.2 Let G be a simple graph. Let $(\mathcal{C}, B, L)$ be its triangle spanning decomposition. Then

$$\alpha(G) = \sum_{C\in\mathcal{C}} \alpha(C) \tag{54}$$

Proof Recall $\alpha(G)$ is the optimal value of

$$\sum_{v\in V(G)} x_v + \sum_{e\in E} y_e \tag{55}$$

subject to $x_v, y_e \geq 0$ and for all triangles $T \in \mathcal{T}(G)$,

$$\sum_{v\in V(T)} x_v + \sum_{e\in E(T)} y_e \geq 1. \tag{56}$$

By Lemma 4.1, no vertex of L lies in triangle, so for every $v \in L$, x_v appears in no constraint. Hence at the optimum we may take $x_v = 0$. By a symmetric argument, $y_e = 0$ for all $e \in B$ at the optimum. Thus $\alpha(G)$ equals the optimal value of

$$\sum_{v\in V(\mathcal{C})} x_v + \sum_{e\in E(\mathcal{C})} y_e \tag{57}$$

subject to the same constraints. By Lemma 4.1, because $V(\mathcal{C})$ and $E(\mathcal{C})$ can be partitioned into $\{V(C)\}_{C\in\mathcal{C}}$ and $\{E(C)\}_{C\in\mathcal{C}}$,

$$\sum_{v\in V(\mathcal{C})} x_v + \sum_{e\in E(\mathcal{C})} y_e = \sum_{C\in\mathcal{C}} \left(\sum_{v\in V(C)} x_v + \sum_{e\in E(C)} y_e \right). \tag{58}$$

Moreover, every triangle $T \in \mathcal{T}(G)$ lives entirely within a unique $C \in \mathcal{C}$. Conversely for all $C \in \mathcal{C}$, every triangle in C is a triangle in G. So the constraint set decouples: the constraints involving variables from a fixed $C \in \mathcal{C}$ are exactly the triangle constraints of $\mathcal{T}(C)$, and no constraint mixes variables from different components. Hence the LP seperates into a sum of n independent sub-LPs, one per component, and

$$\begin{aligned} \alpha(G) &= \sum_{C \in \mathcal{C}} \min \left\{ \sum_{v \in V(C)} x_v + \sum_{e \in E(C)} y_e \;\middle|\; x, y \geq 0, (x, y) \text{ feasible} \right\} \\ &= \sum_{C \in \mathcal{C}} \alpha(C) \end{aligned} \tag{59}$$

□

6. Upper Bound for Triangle Spanning Graphs

In this section, we will prove the upper bound holds for triangle spanning graphs.

Proposition 6.1 Let G be a triangle spanning graph, and let $V = |V(G)|$. Then for all $(e, \tau) \in \mathcal{R}$,

$$T^G_{\max}(\varepsilon, \tau) \leq \tau^{\alpha(G)}. \tag{60}$$

Furthermore, for $T^G_{\max}(\varepsilon, \tau) = O\big(\tau^{\alpha(G)}\big)$ for fixed $\varepsilon \in \big(0, \frac{1}{2}\big]$ and $\tau \to 0$.

Proof Fix any $W \in \widetilde{\mathcal{W}}(\varepsilon, \tau)$ where $(\varepsilon, \tau) \in R$. Let $\{z^*_\tau\}_{\tau \in \mathcal{T}(G)}$ be optimal solution to the dual LP to $\alpha(G)$. By Strong Duality, $\alpha(G) = \sum_{\tau \in \mathcal{T}(G)} z^*_\tau$. For $x \in [0,1]^V$ and for all $T = \{i, j, k\} \in \mathcal{T}(G)$, let

$$x_T := \begin{pmatrix} x_i \\ x_j \\ x_j \end{pmatrix}. \tag{61}$$

Let $W_\triangle : [0,1]^3 \to [0,1]$ where

$$W_\triangle(x_1, x_2, x_3) = W(x_1, x_2) W(x_2, x_3) W(x_1, x_3). \tag{62}$$

By the Dual LP, $\sum_{T \ni e} z^*_e \in [0,1]$ for all edges $e \in E(G)$. Since $W \leq 1$, raising W to a exponent in $[0,1]$ can only increase the value. Thus

$$\prod_{ij \in E(G)} W\big(x_i, x_j\big) \leq \prod_{ij \in E(G)} W\big(x_i, x_j\big)^{\sum_{T \ni ij} z^*_T} = \prod_{T \in \mathcal{T}(G)} W_\triangle(x_T)^{z^*_T}, \tag{63}$$

where the last equality comes from grouping all the edges into their respective triangles. Thus,

$$t(G,W) \leq \int_{[0,1]^V} \prod_{T\in\mathcal{T}(G)} W_\triangle(x_T)^{z_T^*}\,\mathrm{d}\vec{x} \tag{64}$$

For all $T=\{i,j,k\}\in\mathcal{T}(G)$, let $S_T=T$, $p_T=(z_T^*)^{-1}$, and $f_T=W_\triangle(x_T)^{z_T}$. Thus by the Corollary to Finner's Inequality,

$$\begin{aligned}
t(G,W) &\leq \int_{[0,1]^V} \prod_{T\in\mathcal{T}(G)} f_T(x_T)\,\mathrm{d}\vec{x} \\
&\leq \prod_{T\in\mathcal{T}(G)} \|f_T\|_{p_T} \\
&= \prod_{T\in\mathcal{T}(G)} \left(\int_{[0,1]^3} \left(W_\triangle(x_1,x_2,x_3)^{z_T^*}\right)^{p_T}\mathrm{d}\vec{x}\right)^{p_T} \\
&= \prod_{T\in\mathcal{T}(G)} \left(\int_{[0,1]^3} W_\triangle(x_1,x_2,x_3)\,\mathrm{d}\vec{x}\right)^{z_T^*} \\
&= \prod_{T\in\mathcal{T}(G)} t(W)^{z_T^*} \\
&= t(W)^{\sum_{T\in\mathcal{T}(G)} z_T^*} \\
&= t(W)^{\alpha(G)} \leq \tau^{\alpha(G)}
\end{aligned} \tag{65}$$

Since $t(G,W)\leq\tau^{\alpha(G)}$ is true for all $W\in\widetilde{\mathcal{W}}(\varepsilon,\tau)$,

$$T^G_{\max}(\varepsilon,\tau)\leq\tau^{\alpha(G)}. \tag{66}$$

This implies, by definition, $T^G_{\max}(\varepsilon,\tau)=O\big(\tau^{\alpha(G)}\big)$ for fixed $\varepsilon\in\big(0,\frac{1}{2}\big]$ and $\tau\to 0$. □

7. Main Result

Putting all the results together of the last few sections, we can finally prove that the upper bound holds for all simple graphs.

Proposition 7.1 Let G be simple graph, where $V=|V(G)|$, and $(\mathcal{C},B,L)$ is its triangle spanning decomposition. Then for all $(e,\tau)\in\mathcal{R}$,

$$T^G_{\max}(\varepsilon,\tau)\leq\tau^{\alpha(G)}. \tag{67}$$

Furthermore, for $T^G_{\max}(\varepsilon,\tau)=O\big(\tau^{\alpha(G)}\big)$ for fixed $\varepsilon\in\big(0,\frac{1}{2}\big]$ and $\tau\to 0$.

Proof Let W be a graphon where $e(W) = e$ and $t(W) = \tau$. By definition $W \leq 1$, so we have the following for $\vec{x} \in [0,1]^V$

$$\begin{aligned}\prod_{\{i,j\}\in E(G)} W\big(x_i, x_j\big) &\leq \prod_{\{i,j\}\in E(\mathcal{C})} W\big(x_i, x_j\big) \\ &= \prod_{C\in\mathcal{C}} \prod_{\{i,j\}\in E(C)} W\big(x_i, x_j\big),\end{aligned} \tag{68}$$

where the last equality comes from Lemma 4.1. Thus

$$t(G,W) \leq \int_{[0,1]^V} \prod_{C\in\mathcal{C}} \prod_{\{i,j\}\in E(C)} W\big(x_i, x_j\big)\,\mathrm{d}\vec{x}. \tag{69}$$

Again note by Lemma 4.1, for all $v \in L$ their corresponding variable x_v occurs nowhere in the integrand. By the same lemma, since $\mathcal{C}$ forms a partition of $V(\mathcal{C})$ each vertex in $V(\mathcal{C})$ appears in one and only one maximal triangle spanning component. Thus,

$$\begin{aligned}t(G,W) &\leq \int_{[0,1]^{V-|L|}} \prod_{C\in\mathcal{C}} \prod_{\{i,j\}\in E(C)} W\big(x_i, x_j\big)\,\mathrm{d}\vec{x} \\ &= \prod_{C\in\mathcal{C}} \int_{[0,1]^{|V(C)|}} \prod_{\{i,j\}\in E(C)} W\big(x_i, x_j\big)\,\mathrm{d}\vec{x} \\ &= \prod_{C\in\mathcal{C}} t(C,W) \\ &\leq \prod_{C\in\mathcal{C}} T^C_{\max}(\varepsilon,\tau) \leq \prod_{C\in\mathcal{C}} \tau^{\alpha(C_i)}\end{aligned} \tag{70}$$

where the last inequality comes from Proposition 6.1. Thus

$$t(G,W) \leq \tau^{\sum_{C\in\mathcal{C}} \alpha(C)} = \tau^{\alpha(G)} \tag{71}$$

where the equality $\alpha(G) = \sum_{C\in\mathcal{C}} \alpha(C)$ comes from Proposition 5.2. Thus, if for all valid graphons W, $t(G,W) \leq \tau^{\alpha(G)}$ then

$$T^G_{\max}(\varepsilon,\tau) \leq \tau^{\alpha(G)} \tag{72}$$

and

$$T^G_{\max}(\varepsilon,\tau) = O\big(\tau^{\alpha(G)}\big) \tag{73}$$

for fixed $\varepsilon \in \big(0, \frac{1}{2}\big]$ and $\tau \to 0$. □

Having established the upper bound, it remains to show that this bound is tight. We demonstrate this by explicitly constructing a family of graphons $W_\tau \in \widetilde{\mathcal{W}}(\varepsilon, \tau)$ that saturates the upper bound as $\tau \to 0$. We achieve this through a multipodal graphon construction parameterized by the optimal primal LP solutions. Proposition 7.2 does it for $\varepsilon \in \left(0, \frac{1}{2}\right)$, while Proposition 7.3 addresses the boundary case $\varepsilon = \frac{1}{2}$.

Definition 7.1

$$T^G_{\max,N} := \max\Big\{t(G, W) \;\Big|\; W \in \widetilde{\mathcal{W}}(N; \varepsilon, \tau)\Big\} \tag{74}$$

Proposition 7.2 Let G simple graph with $\mathcal{T}(G) \neq \emptyset$. Let $V = |V(G)|$ and $(\mathcal{C}, B, L)$ is its triangle spanning decomposition.

$$T^G_{\max,N}(\varepsilon, \tau) = \Omega\big(\tau^{\alpha(G)}\big) \tag{75}$$

for fixed $\varepsilon \in (0, 0.5)$ and fixed $N \geq V + 2$ and $\tau \to 0$.

Proof Let $\varepsilon \in (0, 0.5)$ be fixed and let $\tau > 0$ be the parameter in $T^G_{\max,N}(\varepsilon, \tau)$. Let $\gamma = \tau/6|\mathcal{T}(G)|$. Let $n = |\mathcal{C}|$, thus $\mathcal{C} = \{C_1, ..., C_n\}$ through a bijection with $[n]$. Let $V_C = |V(\mathcal{C})|$ and $V_L = |L|$.

Let B_0, B_1, B_2 be a partition of B where B_0 is the L-to-L edges, B_1 are the L-to-$V(\mathcal{C})$ edges, and B_2 are the $V(\mathcal{C})$-to-$V(\mathcal{C})$ edges.

Let (x^*, y^*) be optimal for $\alpha(G)$, with the convention that $x^*(L) = \{0\}$ and $y^*(B) = \{0\}$ (optimal, since these variables appear in no triangle constraint).

Fix any $a_0 \in \left(\sqrt{\varepsilon/2}, 0.5\right)$; this interval is nonempty because $\varepsilon < 0.5$.

Let $\nu \in (0, \nu_{\max}]$ where

$$\nu_{\max} := \min\left(\frac{1 - 2a_0}{V_L}, \sqrt{\frac{\varepsilon}{2|B_0| + 1}}\right). \tag{76}$$

Let

$$\sigma(\gamma) := \sum_{v \in V(\mathcal{C})} \gamma^{x_i^*}. \tag{77}$$

Note as $\gamma \to 0$, $\sigma(\gamma) \to 0$.

Let $\{S_i\}_{i=1}^N$ be pairwise disjoint measurable sets in $[0, 1]$ where:

$$\mu(S_i) = \begin{cases} \gamma^{x_i^*} \text{ if } i \in V(\mathcal{C}) \\ \nu \text{ if } i \in L \\ a(N) \text{ if } i \in \{V+1, V+2\} \\ b(\gamma) \text{ otherwise} \end{cases} \tag{78}$$

where

$$\begin{aligned} a(N) &:= \begin{cases} a_0 + \frac{1}{2}(1 - 2a_0 - \sigma(\gamma) - V_L\nu) \text{ if } N = V+2 \\ a_0 \text{ otherwise} \end{cases} \\ b(\gamma) &:= \frac{1 - 2a_0 - \sigma(\gamma) - V_L\nu}{N - V - 2} \text{ when } (N \geq V+2) \end{aligned} \tag{79}$$

Since $\gamma \to 0 \Rightarrow \sigma(\gamma) \to 0$ and $2a_0 + V_L\nu > 1$, there exists a $\gamma_0 > 0$ such that $b(\gamma) > 0$ for all $\gamma \in (0, \gamma_0]$, and thus a valid measure. Furthermore, by construction

$$\sum_{i=1}^{N} \mu(S_i) = 1 \tag{80}$$

So $\{S_i\}_{i=1}^N$ is a partition of $[0, 1]$.

Let

$$\begin{aligned} R(\gamma) &:= 2 \sum_{ij \in E(\mathcal{C})} \gamma^{x_i^* + x_j^* + y_{ij}^*}, \\ Q_0 &:= 2 \sum_{e \in B_0} \nu^2 = 2|B_0|\nu^2, \\ Q_1(\gamma) &:= 2 \sum_{\{i,j\} \in B_1} \nu\gamma^{x_{\text{non}-V_0}^*} \\ Q_2(\gamma) &:= 2 \sum_{\{i,j\} \in B_2} \gamma^{y_{ij}^*} \\ \beta(\gamma) &= \frac{\varepsilon - R(\gamma) - Q_0 - Q_1(\gamma) - Q_2(\gamma)}{2\mu(S_{V+1})\mu(S_{V+2})} \end{aligned} \tag{81}$$

Let $w : [N]^2 \to [0, 1]$ where

$$w_{ij} = \begin{cases} \gamma^{y_{ij}^*} \text{ if } \{i, j\} \in E(\mathcal{C}) \\ 1 \text{ if } \{i, j\} \in B \\ \beta(\gamma) \text{ if } \{i, j\} = \{V+1, V+2\} \\ 0 \text{ otherwise} \end{cases}. \tag{82}$$

As $\gamma \to 0$,

$$\beta(\gamma) \to \frac{\varepsilon - Q_0}{2a_0^2} = \frac{\varepsilon - 2|B_0|\nu^2}{2a_0^2} > 0. \tag{83}$$

Since $a_0^2 > \varepsilon/2$, the limit is less than 1. Hence there exists a $\gamma_1 \in (0, \gamma_0]$ such that $\beta(\gamma) \in (0,1)$ for all $\gamma \in (0, \gamma_1)$.

Let W be an N-podal graphon with parts $\{S_i\}_{i=1}^N$ and edge weights w_{ij}.

$$\begin{aligned} e(W) &= \sum_{i,j\in[N]} \mu(S_i)\mu(S_j)w_{ij} \\ &= 2\beta(\gamma)\mu(S_{V+1})\mu(S_{V+2}) + R(\gamma) + Q_0 + Q_1(\gamma) + Q_2(\gamma) \\ &= \varepsilon \end{aligned} \tag{84}$$

Thus W has the correct edge density. For the triangle density,

$$t(W_0) = \sum_{i,j,k\in[N]} \mu(S_i)\mu(S_j)\mu(S_k)w_{ij}w_{ik}w_{jk}. \tag{85}$$

The only way that w_{ij}, w_{ik}, w_{jk} are all nonzero is if $\{i,j,k\} \in \mathcal{T}(G)$. Triangles of G are contained in components $C_l \in \mathcal{C}$, so all three edges must be in $E(C_l) \subset E(\mathcal{C}) \subset$. No edge in B may participate. Thus

$$\begin{aligned} t(W_0) &= 6 \sum_{\{i,j,k\}\in\mathcal{T}(G)} \mu(S_i)\mu(S_j)\mu(S_k)w_{ij}w_{ik}w_{jk} \\ &= 6 \sum_{\{i,j,k\}\in\mathcal{T}(G)} \gamma^{x_i+x_j+x_k}\gamma^{y_{ij}+y_{jk}+y_{ik}} \\ &= 6 \sum_{T\in\mathcal{T}(G)} \gamma^{\sum_{i\in V(T)} x_i + \sum_{e\in E(T)} y_e} \\ &= 6 \sum_{T\in\mathcal{T}(G)} \gamma^{\sum_{i\in V(T)} x_i + \sum_{e\in E(T)} y_e} \end{aligned} \tag{86}$$

By the feasibility constraint on $\alpha(G)$, $\forall T \in \mathcal{T}$

$$\sum_{i\in V(T)} x_i^* + \sum_{e\in E(T)} y_e^* \geq 1. \tag{87}$$

Since $\gamma \to 0 \Rightarrow \gamma < 1$, that means for all $T \in \mathcal{T}$,

$$\gamma^{\sum_{i\in V(T)} x_i + \sum_{e\in E(T)} y_e} \leq \gamma^1 \tag{88}$$

Thus

$$t(W_0) \leq 6|\mathcal{T}(G)|\gamma = \tau. \tag{89}$$

Furthermore by Lemma 5.1, there exists a $T_0 \in \mathcal{T}(G)$ such that

$$\sum_{v \in V(T_0)} x_v + \sum_{e \in E(T_0)} y_e = 1. \tag{90}$$

Since all terms in the sum of $t(W)$ are positive, $t(W) \geq 6\tau = |\mathcal{T}(G)|^{-1}\tau$. Thus $t(W) = \Theta(\tau)$.

Now for $t(G, W)$:

$$\begin{aligned} t(G, W) &= \sum_{\varphi: V(G) \to [N]} \Pi_\varphi \\ \Pi_\varphi &= \left(\prod_{v \in V(G)} \mu\big(S_{\varphi(v)}\big) \right) \left(\prod_{e \in E(G)} w_{\varphi(e)} \right) \end{aligned} \tag{91}$$

For Π_φ to be nonzero, every edge $e \in E(G)$ must satisfy $w_{\varphi(e)} \neq 0$, meaning $\varphi(e) \in E(G)$ or $\varphi(e) \in \{V+1, V+2\}$.

Suppose for contradiction some $e = \{u, v\} \in E(G)$ has $\varphi(e) = \{V+1, V+2\}$. Without loss of generality, $\varphi(u) = V+1$. Since G is triangle spanning, u lies in some triangle $T = \{u, v, w\} \in \mathcal{T}(G)$ with edges e, $f = \{u, w\}$, and $g = \{v, w\}$ in $E(G)$. For $\Pi_\varphi \neq 0$, we need $w_{\varphi(f)} \neq 0$. Since $\varphi(u) = V+1$ and the only k such that $w_{V+1,k} \neq 0$ is $k = V+2$, we must have $\varphi(w) = V+2$. Symmetrically, $w_{\varphi(g)} \neq 0$ forces $\varphi(w) = V+1$. These are incompatible, so $\Pi_\varphi \neq 0$.

Hence every φ where Π_φ satisfies $\varphi(E(G)) \subset E(G)$, $\varphi(V(G)) \subset [V]$. So,

$$t(G, W_0) = \sum_{\varphi: V(G) \to [V]} \Pi_\varphi \tag{92}$$

Take $\varphi = \mathrm{id}$: The identity map satisfies $\mathrm{id}(E(G)) \subset E(G)$ so it contributes a nonzero term.

$$\begin{aligned} \Pi_{\mathrm{id}} &= \left(\prod_{v \in V(G)} \mu(S_v) \right) \left(\prod_{e \in E(G)} w_e \right) \\ &= \left(\prod_{v \in V(G)} \gamma^{x_v^*} \right) \left(\prod_{e \in E(G)} \gamma^{y_e^*} \right) \\ &= \gamma^{\sum_{v \in V(G)} x_v^* + \sum_{e \in E(G)} y_e^*} = \gamma^{\alpha(G)} \end{aligned} \tag{93}$$

Since all $\Pi_\varphi \geq 0$,

$$t(G,W) \geq \Pi_{\text{id}} = \gamma^{\alpha(G)} = \left(\frac{\tau}{6|\mathcal{T}(G)|}\right)^{\alpha(G)} = \frac{1}{(6|\mathcal{T}(G)|)^{\alpha(G)}}\tau^{\alpha(G)} \tag{94}$$

Since W is a valid N-podal graphon with $e(W) = \varepsilon$ and $t(W) \leq \tau$, so W is feasible in the definition of $T^G_{\max,N}(\varepsilon,\tau)$. Therefore,

$$T^G_{\max,N}(e,\tau) \geq t(G,W) \geq \frac{1}{(6|\mathcal{T}(G)|)^{\alpha(G)}}\tau^{\alpha(G)}, \tag{95}$$

giving $T^G_{\max,N}(\varepsilon,\tau) = \Omega\big(\tau^{\alpha(G)}\big)$ as $\tau \to 0$. □

Proposition 7.3 Let G simple graph with $\mathcal{T}(G) \neq \emptyset$. Let $V = |V(G)|$ and $(\mathcal{C}, B, L)$ is its triangle spanning decomposition.

$$T^G_{\max,V+2}\left(\frac{1}{2},\tau\right) = \Omega\big(\tau^{\alpha(G)}\big) \tag{96}$$

for $\tau \to 0$.

Proof Let $\tau > 0$ be the parameter in $T^G_{\max,N}\big(\frac{1}{2},\tau\big)$. Let $\gamma = \tau/6|\mathcal{T}(G)|$. Let $n = |\mathcal{C}|$, thus $\mathcal{C} = \{C_1, ..., C_n\}$ through a bijection with $[n]$. Let $V_C = |V(\mathcal{C})|$ and $V_L = |L|$.

Let B_0, B_1, B_2 be a partition of B where B_0 is the L-to-L edges, B_1 are the L-to-$V(\mathcal{C})$ edges, and B_2 are the $V(\mathcal{C})$-to-$V(\mathcal{C})$ edges.

Let (x^*, y^*) be optimal for $\alpha(G)$, with the convention that $x^*(L) = \{0\}$ and $y^*(B) = \{0\}$ (optimal, since these variables appear in no triangle constraint).

Let $\nu \in (0, \nu_{\max}]$ where

$$\nu_{\max} := \min\left(\frac{1}{V_L+1}, \sqrt{\frac{1}{4|B_0|+1}}\right). \tag{97}$$

Let

$$\sigma(\gamma) := \sum_{v \in V(\mathcal{C})} \gamma^{x_i^*}. \tag{98}$$

Note as $\gamma \to 0$, $\sigma(\gamma) \to 0$.

Let $\{S_i\}_{i=1}^N$ be pairwise disjoint measurable sets in $[0,1]$ where:

$$\mu(S_i) = \begin{cases} \gamma^{x_i^*} \text{ if } i \in V(\mathcal{C}) \\ \nu \text{ if } i \in L \\ a(\gamma) \text{ if } i \in \{V+1, V+2\} \end{cases} \tag{99}$$

where

$$a(\gamma) := \frac{1}{2}(1 - V_L\nu - \sigma(\gamma)) \tag{100}$$

Since $\gamma \to 0 \Rightarrow \sigma(\gamma) \to 0$ and $1 - V_L\nu > 0$, there exists a $\gamma_0 > 0$ such that $b(\gamma) > 0$ for all $\gamma \in (0, \gamma_0]$, and thus a valid measure. Furthermore, by construction

$$\sum_{i=1}^{N} \mu(S_i) = 1 \tag{101}$$

So $\{S_i\}_{i=1}^N$ is a partition of $[0, 1]$. Let

$$\begin{aligned}
R(\gamma) &:= 2 \sum_{ij \in E(\mathcal{C})} \gamma^{x_i^* + x_j^* + y_{ij}^*}, \\
Q_0 &:= 2 \sum_{e \in B_0} \nu^2 = 2|B_0|\nu^2, \\
Q_1(\gamma) &:= 2 \sum_{\{i,j\} \in B_1} \nu\gamma^{x_{\text{non}-V_0}^*} \\
Q_2(\gamma) &:= 2 \sum_{\{i,j\} \in B_2} \gamma^{y_{ij}^*} \\
\beta(\gamma) &= \frac{0.5 - R(\gamma) - Q_0 - Q_1(\gamma) - Q_2(\gamma)}{2\mu(S_{V+1})\mu(S_{V+2})}
\end{aligned} \tag{102}$$

Let $w : [N]^2 \to [0, 1]$ where

$$w_{ij} = \begin{cases} \gamma^{y_{ij}^*} \text{ if } \{i, j\} \in E(\mathcal{C}) \\ 1 \text{ if } \{i, j\} \in B \\ \beta(\gamma) \text{ if } \{i, j\} = \{V + 1, V + 2\} \end{cases}. \tag{103}$$

As $\gamma \to 0$,

$$\beta(\gamma) \to \frac{0.5 - Q_0}{2a_0^2} = \frac{0.5 - 2|B_0|\nu^2}{2a_0^2} > 0. \tag{104}$$

Since $a_0^2 > \varepsilon/2$, the limit is less than 1. Hence there exists a $\gamma_1 \in (0, \gamma_0]$ such that $\beta(\gamma) \in (0, 1)$ for all $\gamma \in (0, \gamma_1)$.

Let W be an N-podal graphon with parts $\{S_i\}_{i=1}^N$ and edge weights w_{ij}.

$$
\begin{aligned}
e(W) &= \sum_{i,j\in[N]} \mu(S_i)\mu(S_j)w_{ij} \\
&= 2\beta(\gamma)\mu(S_{V+1})\mu(S_{V+2}) + R(\gamma) + Q_0 + Q_1(\gamma) + Q_2(\gamma) \\
&= 0.5
\end{aligned} \tag{105}
$$

Thus W has the correct edge density. For $t(W) \leq \tau$ and

$$
t(G, W) \geq \frac{1}{(6|\mathcal{T}(G)|)^{\alpha(G)}} \tau^{\alpha(G)}, \tag{106}
$$

the proof follows identically to Proposition 7.2. Since W is a valid $(V+2)$-podal graphon with $e(W) = 0.5$ and $t(W) \leq \tau$, so W is feasible in the definition of $T^G_{\max,V+2}(\varepsilon, \tau)$. Therefore,

$$
T^G_{\max,V+2}\left(\frac{1}{2}, \tau\right) \geq t(G, W) \geq \frac{1}{(6|\mathcal{T}(G)|)^{\alpha(G)}} \tau^{\alpha(G)}, \tag{107}
$$

giving $T^G_{\max,V+2}\left(\frac{1}{2}, \tau\right) = \Omega\left(\tau^{\alpha(G)}\right)$ as $\tau \to 0$. □

Theorem 7.4 G be a simple graph where $\mathcal{T}(G) \neq 0$.

$$
T^G_{\max}(e, \tau) = \Theta\left(\tau^{\alpha(G)}\right) \tag{108}
$$

for $.e \in \left(0, \frac{1}{2}\right]$ and $\tau \to 0$.

Proof By Proposition 7.2 and Proposition 7.3, for $V = |V(G)|$

$$
T^G_{\max,V+2}(\varepsilon, \tau) \geq c_1 \tau^{\alpha(G)}. \tag{109}
$$

Since

$$
T^G_{\max,V+2}(\varepsilon, \tau) \leq T^G_{\max}(\varepsilon, \tau) \Rightarrow c_1 \tau^{\alpha(G)} \leq T^G_{\max}(\varepsilon, \tau). \tag{110}
$$

Combining this with Proposition 7.1, we have

$$
T^G_{\max}(\varepsilon, \tau) = \Theta\left(\tau^{\alpha(G)}\right) \tag{111}
$$

□

8. Examples

9. Conclusion and Future Directions

This thesis identifies the exponent governing the asymptotic behavior of $T^G_{\max}(\varepsilon, \tau)$ as $\tau \to 0$ for $\varepsilon \in \left(0, \frac{1}{2}\right]$: for every simple graph G, $T^G_{\max}(\varepsilon, \tau) = \Theta\left(\tau^{\alpha(G)}\right)$, where $\alpha(G)$ is the optimal value of an explicit linear program indexed by the triangles of G. The upper bound is obtained by combining Finner's generalized Hölder inequality with strong duality applied $\alpha(G)$. The lower bound is obtained by an explicit step-graphon construction guided by the LP optimizer. Several natural questions remain open, and we conclude by describing three directions in which the present work suggests further investigation.

9.1. Sharpening the constants

The bounds in this thesis determine the constant $\alpha(G)$ exactly, but leave a room between the upper and lower implicit constants. The upper bound, obtained by Finner's Inequality, gives $T^G_{\max}(\varepsilon, \tau) \leq \tau^{\alpha(G)}$. The lower bound construction gives $T^G_{\max}(\varepsilon, \tau) \geq (6|\mathcal{T}(G)|)^{-\alpha(G)}\tau^{\alpha(G)}$. The $(6|\mathcal{T}(G)|)^{\alpha(G)}$ is already significant for small graphs and quickly grows as the complexity of the graph grows, so there is a significant gap between the upper and lower bound constants.

The quantity of interest is the true asymptotic constant,

$$C(G, \varepsilon) := \lim_{\tau \to 0} \frac{T^G_{\max}(\varepsilon, \tau)}{\tau^{\alpha(G)}}, \tag{112}$$

when the limit exists, or the associated lim sup or lim inf otherwise. Identifying $C(G, \varepsilon)$ (or even showing the limit exists) would involve solving a variational problem to find the extremizing family of graphons as $\tau \to 0$, not merely identifying the scaling exponent. Neither bound in this thesis is likely tight:

- Upper Bound: Finner's Inequality only achieves equality when each factor decomposes into a product of single variable functions, which fails for nontrivial graphons.
- Lower Bound: The multipodal construction is specific enough that a more refined family of extremizers likely exists.

One also suspects that $C(G, \varepsilon)$ has a interesting dependence on the edge density, which the current bounds don't detect.

9.2. Behavior in other regimes of the Razborov triangle.

The results of this thesis pin down the asymptotic exponent of $T^G_{\max}(\varepsilon, \tau)$ only on the lower boundary of the Razborov triangle, namely as $\tau \to 0$ and $\varepsilon \in \left(0, \frac{1}{2}\right]$ fixed. Elsewhere in the feasible region, the scaling is less well understood. By Proposition 7.1, we know that $T^G_{\max}(\varepsilon, \tau) \leq \tau^{\alpha(G)}$ for all $(\varepsilon, \tau) \in \mathcal{R}$. The matching lower bound, however, is only proven for

$\varepsilon \in \left(0, \frac{1}{2}\right]$ as $\tau \to 0$. In other regimes of the Razborov triangle, the exponent governing $T^G_{\max}$ is not understood. Furthermore, in some cases, it is almost certainly not $\alpha(G)$.

9.2.1. Near the Upper Boundary $\tau = \varepsilon^{\frac{3}{2}}$

The extremizer of the triangle density is the unique graphon $W = \mathbb{1}_{[0,\sqrt{\varepsilon}]^2}$ (Theorem 2.7), for which $t(G, W) = \varepsilon^{|E(G)|}$. Hence $T^G_{\max}\left(\varepsilon, \varepsilon^{\frac{3}{2}}\right) = \varepsilon^{|E(G)|}$, which can be compared to the upper bound $\varepsilon^{3\alpha(G)/2}$ evaluated at the boundary. For most G the two values differ, and since the upper bound is not tight at this corner, the true scaling $T^G_{\max}(\varepsilon, \tau)$ as $\tau \to \varepsilon^{\frac{3}{2}}$ from below is governed by a different exponent than $\alpha(G)$. For this purpose the better function to study is

$$T^G_{\text{gap}}(\varepsilon, \eta) = T^G_{\max}\left(\varepsilon, \varepsilon^{\frac{3}{2}} - \eta\right) - \varepsilon^{|E(G)|}. \tag{113}$$

9.2.2. Near lower boundary of $\mathcal{R}$ as $\varepsilon > \frac{1}{2}$

Here the triangle free region disappears: for edge density above $\frac{1}{2}$, every graphon has strictly positive triangle density, with minimum given by Theorem 2.8, achieved by the constructions of Theorem 2.9. The construction resemble blow-ups of complete k-partite graphs with one of the parts, which will be called the special part, also having a triangle free internal structure. Starting from such a minimum-triangle graphon, adding an infinitesimal amount of extra triangle density cooresponds to adding edges to the special part to form triangles. This doesn't just cause G to pop out from within the special part, but we may get the lions share of the G-density from the interactions of the special part with the rest of the $k-1$ parts. Thus for every triangle we create we may create many copies of G. Again like the case with the upper boundary, the $T^G_{\max}(\varepsilon, t_{\min}(\varepsilon))$ term will dominate $T^G_{\max}(\varepsilon, \tau)$. Thus the function to study is,

$$T^G_{\text{gap}}(\varepsilon, \eta) = T^G_{\max}(\varepsilon, t_{\min}(\varepsilon) + \eta) - T^G_{\max}(\varepsilon, t_{\min}(\varepsilon)). \tag{114}$$

9.3. The opposite problem $T^G_{\min}$ and the Triangle Removal Lemma

A natural companion to the problem studied in this thesis is the opposite extremal question:

$$T^G_{\min}(\varepsilon, \tau) := \min\left\{t(G, W) \;\middle|\; W \in \widetilde{\mathcal{W}}(\varepsilon, \tau)\right\} \tag{115}$$

This is well-posed by the same compactness argument used for $T^G_{\max}$, and understanding its behavior as $\tau \to 0$ is open.

For the case $G = D$ (the diamond graph or K_4 with one edge removed), $T^D_{\min}$ is of particular interest because of its connection to the Szemerédi-Ruzsa Triangle Removal Lemma. This lemma asserts that any graph with few triangles can be made triangle-free by removing a correspondingly small fraction of edges. In the language of the graphons, this is governed by the function:

$$c(\tau) = \sup_{W:t(W)\leq\tau} \left(\inf_{\substack{U\leq W \\ t(U)=0}} \|W-U\|_1 \right), \tag{116}$$

intuitively this function quantifies the maximum edge mass we have to remove from a graphon with triangle density at most τ to guarentee triangle-freeness. Historically, proofs relying on Szemerédi's Regularity Lemma yield bounds for $c(\tau)$ that decay at an inverse tower-exponential rate as $\tau \to 0$ [15].

A simple variational argument based on Cauchy-Schwarz applied to functional derivatives of $t(D,\cdot)$ and $t(\cdot)$ on triangle free graphons, connects understanding $c(\tau)$ to the problem of understanding $T^D_{\min}$, and we get the following inequality

$$t(D,W) \geq \frac{t(W)^2}{9c(t(W))} \text{ near } \tau = 0. \tag{117}$$

Thus

$$T^D_{\min}(\varepsilon,\tau) \geq \frac{\tau^2}{9c(\tau)} \Rightarrow c(\tau) \geq \frac{\tau^2}{9T^D_{\min}(\varepsilon,\tau)} \tag{118}$$

Thus understanding the order of $T^D_{\min}(\varepsilon,\tau)$ for fixed ε and $\tau \to 0$, will give us information on the asymptotics of $c(\tau)$. Understanding the asymptotics of $c(\tau)$ is major a topic of study in extremal combinatorics currently.

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